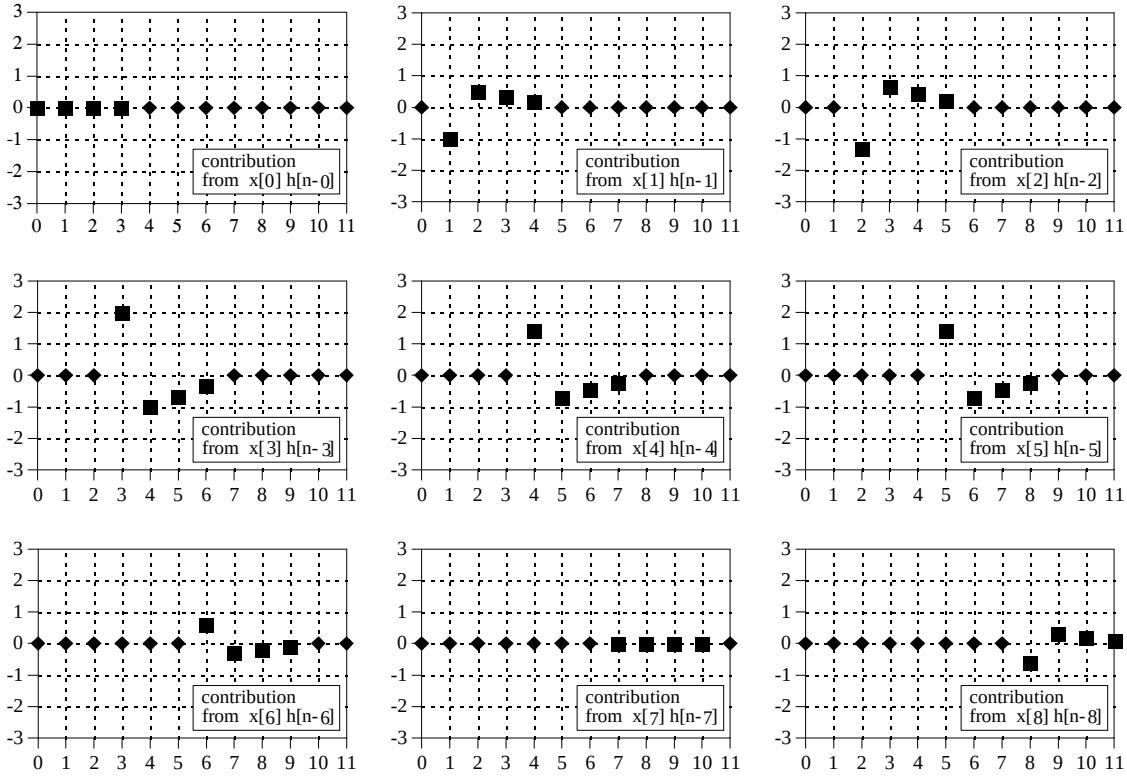

FIGURE 6-6
Output signal components for the convolution in Fig. 6-5. In these signals, each point that results from a scaled and shifted impulse response is represented by a square marker. The remaining data points, represented by diamonds, are zeros that have been added as place holders.

In this example, $x[n]$ is a nine point signal and $h[n]$ is a four point signal. In our next example, shown in Fig. 6-7, we will reverse the situation by making $x[n]$ a four point signal, and $h[n]$ a nine point signal. The same two waveforms are used, they are just swapped. As shown by the output signal components, the four samples in $x[n]$ result in four shifted and scaled versions of the nine point impulse response. Just as before, leading and trailing zeros are added as place holders.

But wait just one moment! The output signal in Fig. 6-7 is *identical* to the output signal in Fig. 6-5. This isn't a mistake, but an important property. Convolution is *commutative*: $a[n] * b[n] = b[n] * a[n]$. The mathematics does not care which is the input signal and which is the impulse response, only that *two signals are convolved with each other*. Although the mathematics may allow it, exchanging the two signals has no physical meaning in system theory. The input signal and impulse response are two totally different things and exchanging them doesn't make sense. What the commutative property provides is a *mathematical tool* for manipulating equations to achieve various results.